Data Structures and Algorithms · 21CSC201J

Unit 2 · Session 11 · SLO 2 — Solved Worksheet

Name: DEEP SAND DEEP SAND | **Reg. No:** RA2511026010557

Unit â€“ 2 : Session â€“ 11 : SLO â€“ 2 | Doubly Linked List â€” Deletion at the Middle

I. Delete a Node After a Given Node NUM

- Step 1: IF START = NULL
- Write UNDERFLOW
- Go to Step 9
- [END OF IF]
- Step 2: SET PTR = START
- Step 3: Repeat Step 4 while PTR -> DATA != NUM
- Step 4: SET PTR = PTR -> NEXT
- [END OF LOOP]
- Step 5: SET TEMP = PTR -> NEXT
- Step 6: SET PTR -> NEXT = TEMP -> NEXT
- Step 7: SET TEMP -> NEXT -> PREV = PTR
- Step 8: FREE TEMP
- Step 9: EXIT

II. Delete a Node Before a Given Node NUM

- Step 1: IF START = NULL
- Write UNDERFLOW
- Go to Step 9
- [END OF IF]
- Step 2: SET PTR = START
- Step 3: Repeat Step 4 while PTR -> DATA != NUM
- Step 4: SET PTR = PTR -> NEXT
- [END OF LOOP]
- Step 5: SET TEMP = PTR -> PREV
- Step 6: SET TEMP -> PREV -> NEXT = PTR
- Step 7: SET PTR -> PREV = TEMP -> PREV
- Step 8: FREE TEMP
- Step 9: EXIT